

TA-212 LAB

PROJECT REPORT, 1st Semester 2025-26

AUTOMATIC BOOK DIGITALISER



GROUP NO. – 25

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Prince 240793

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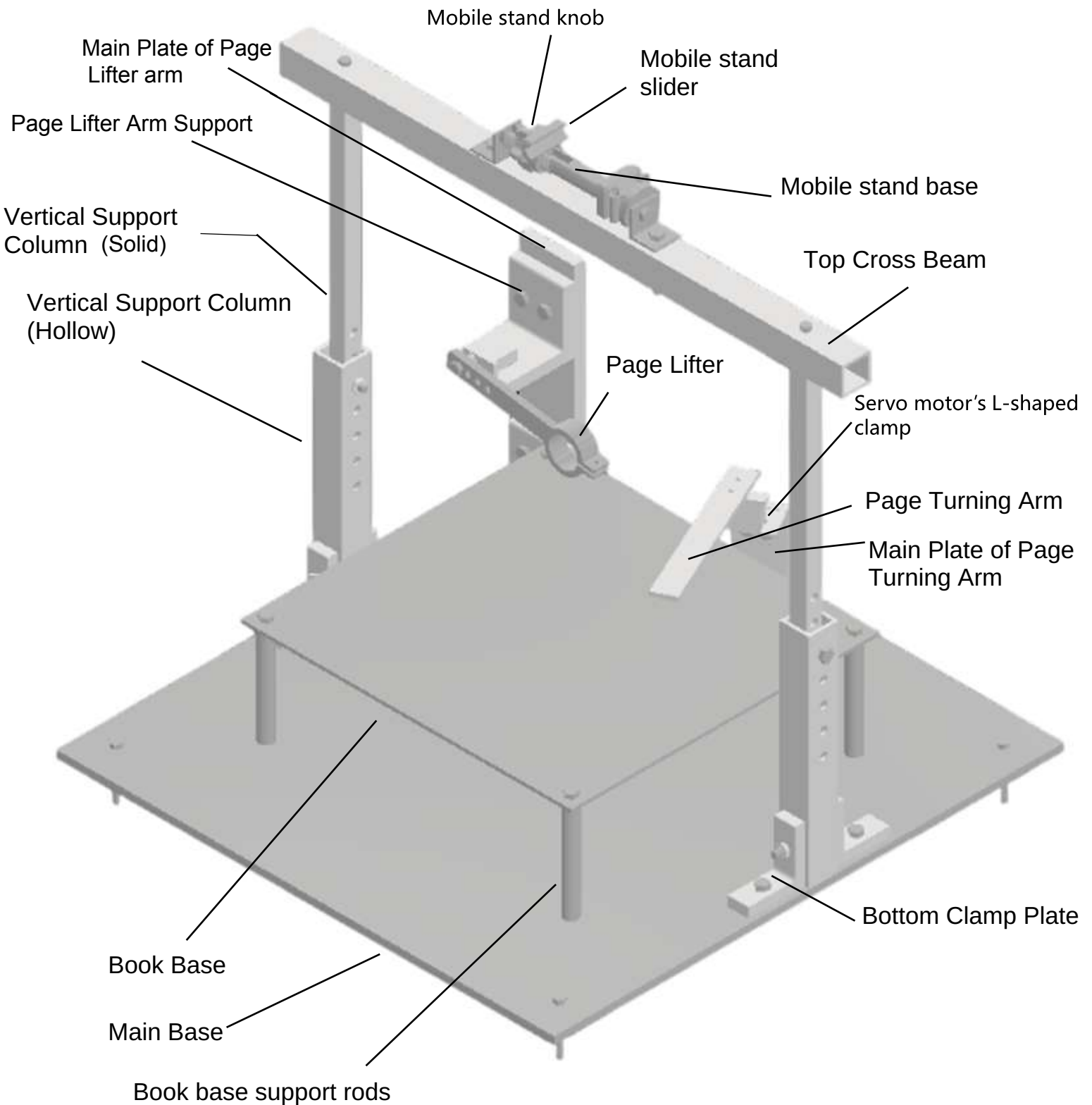
Sheth Namya Jigar 240971

Shivanshu Jaiswara 240985

Tejavath Srinivas 241098

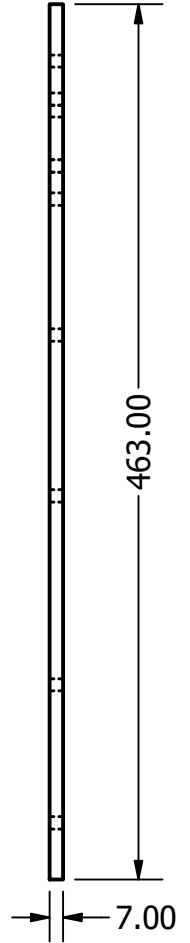
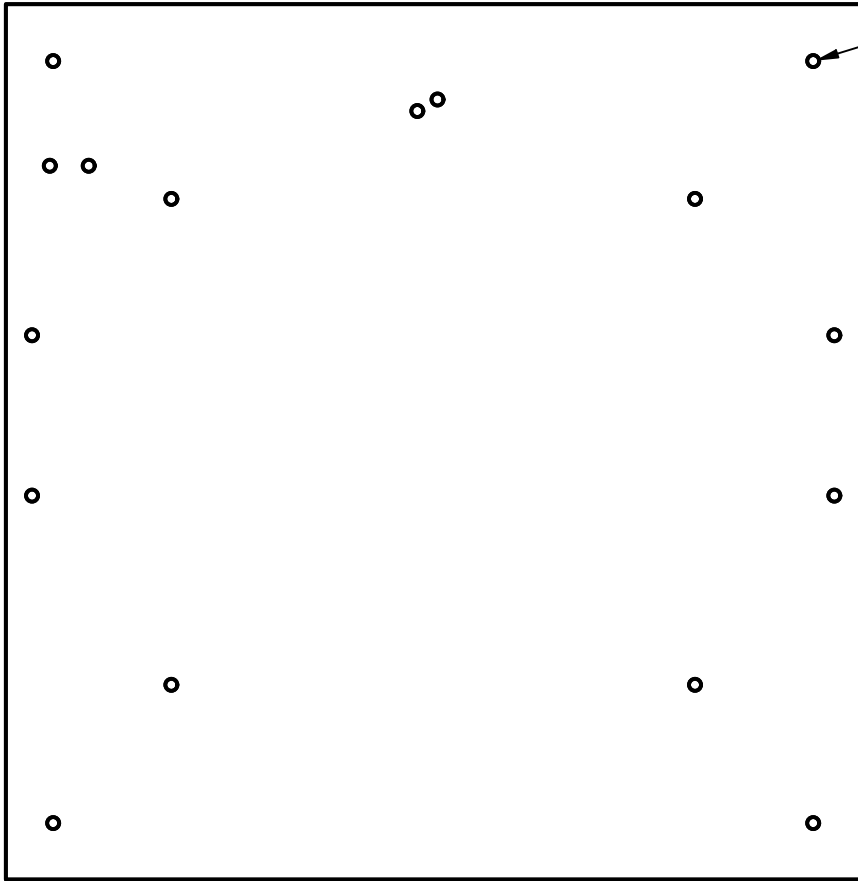
AUTOMATIC BOOK DIGITALISER

Full Assembly Isometric View

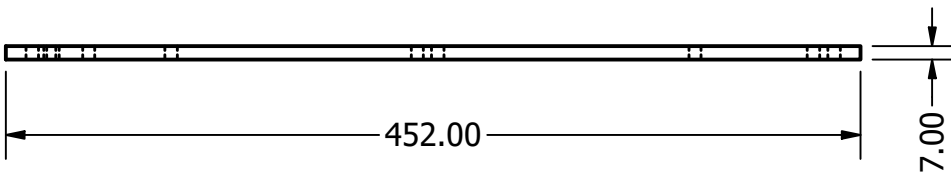


INDEX

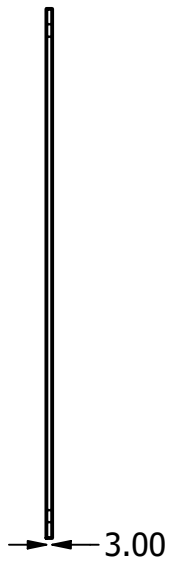
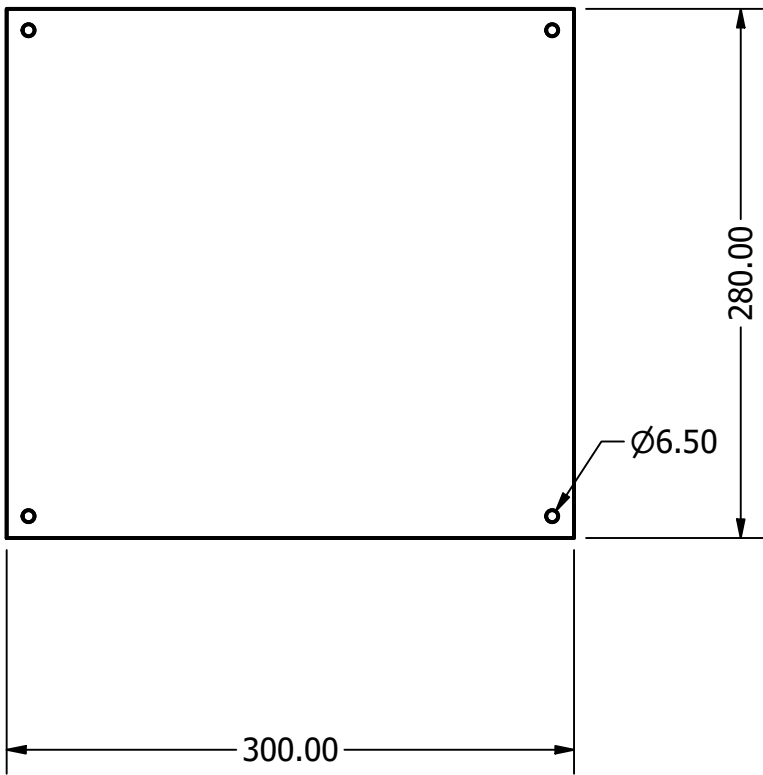
PART NO	PART NAME	QTY	DIMENSIONS (in mm)	MATERIAL	PAGE NO
1	Isometric Drawing	1	-	-	1
2	Main base	1	463 X 452 X 7	Mild Steel	3
3	Book base	1	300 x 280 x 3	Mild Steel	4
4	Book base support rods	4	L = 90.5 D= 16.5	Mild Steel	5
5	Vertical Support Column (Solid)	2	15 x 15 x 237	Mild Steel	6
6	Vertical Support Column (Hollow)	2	27 x 27 x 198	Mild Steel	7
7	Top Cross Beam	1	26 x 26 x 510	Mild Steel	8
8	Bottom Clamp Plate	4	51 x 19 x 51	Mild Steel	9
9	Main Plate of Page Lifter arm	1	50 x 10 x 204	Mild Steel	10
10	Page Lifter Arm Support	1	60 x 50 x 62	Mild Steel	11
11	Page Lifter	1	3D print as per drawing	PLA	12
12	Mobile stand base	1	3D print as per drawing	PLA	13
13	Mobile stand knob	1	3D print as per drawing	PLA	14
14	Mobile stand slider	1	3D print as per drawing	PLA	15
15	Main Plate of Page Turning Arm	1	37.20 x 6.40 x 87.60	Mild Steel	16
16	Servo motor's L-shaped clamp	1	27.30 x 3.10 x 37.20	Mild Steel	17
17	Page Turning Arm	1	3D print as per drawing	PLA	18
18	Nut and Bolt		-	Mild Steel	19
19	Code Implementation	-	-	-	20



Scale 1:2



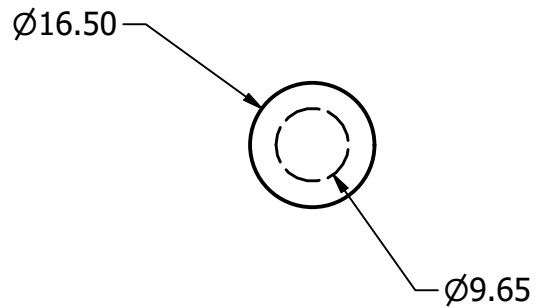
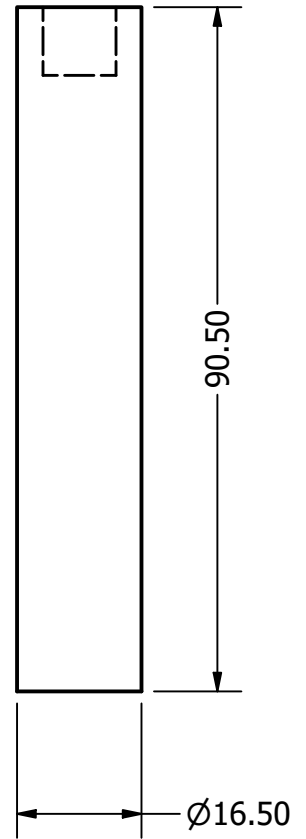
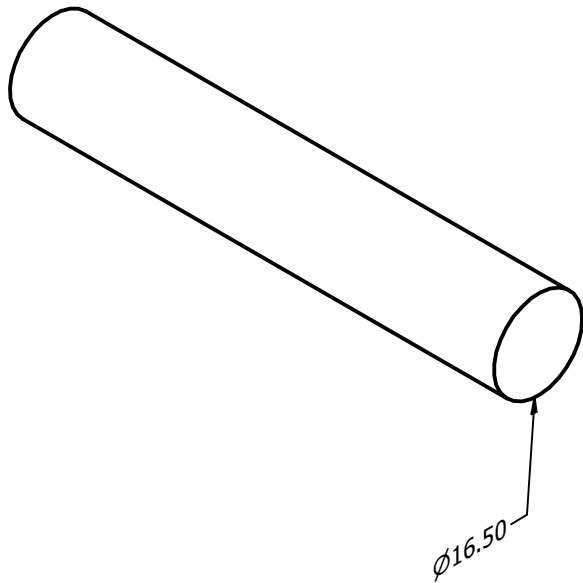
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IK					
CHECKED					
QA					
MFG		TITLE Shivani and Namrata Main Base			
APPROVED					
		SIZE	A4		DWG NO
					REV
		SCALE	1 / 4		
			SHEET 1 OF 1		



Scale 1:1

All dimensions in mm.
Tolerance of dimensions is 0.5mm
Material Required - Mild steel

DRAWN	12-11-2025	TITLE BOOK BASE Shivani and Namrata			
IK					
CHECKED					
QA					
MFG		SIZE A4			
APPROVED					
		SCALE	1 / 4	DWG NO	REV
		SHEET 1 OF 1			

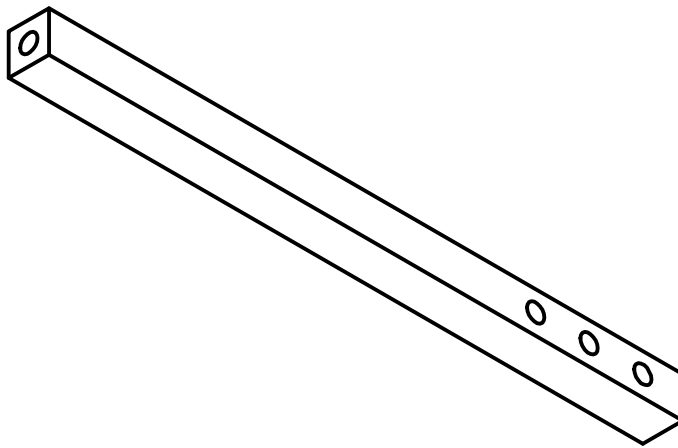


Scale 2:1

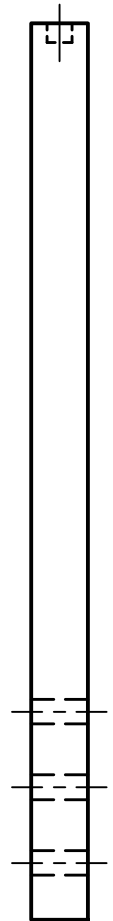
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Materials Required - Mild Steel

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CHECKED						
QA						
MFG						
APPROVED			SIZE	DWG NO		REV
			A4			
			SCALE	1 : 1	SHEET 1 OF 1	

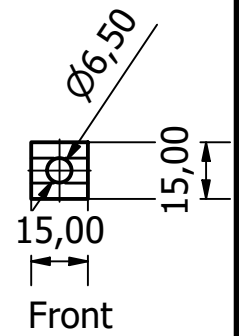
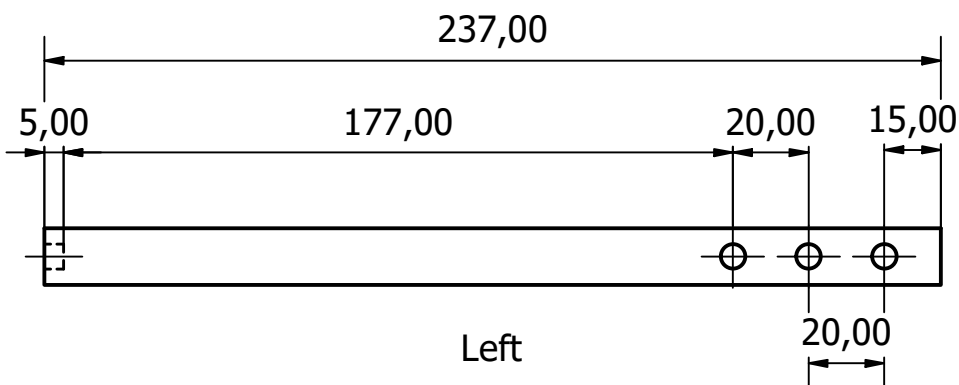
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Isometric

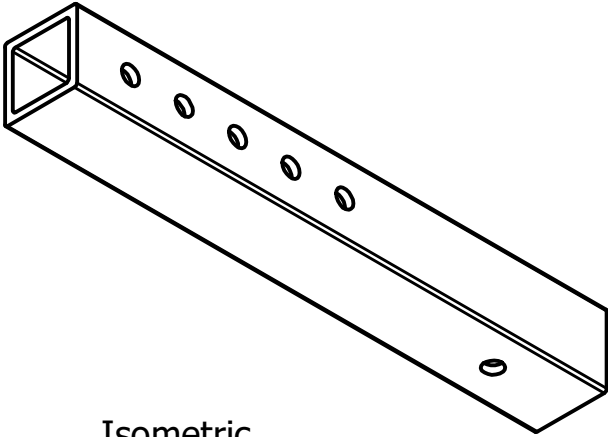


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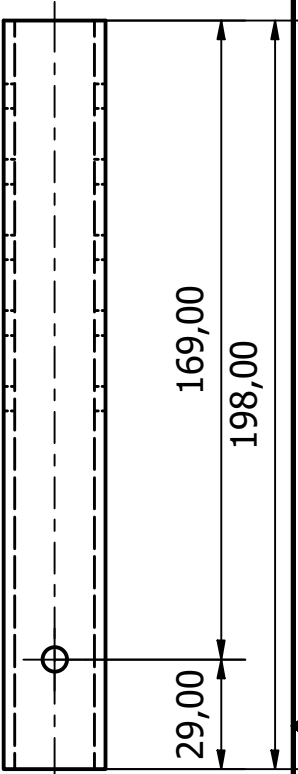


Designed by lover aj	Checked by	Approved by	Date		Date 29-10-2025	
			Vertical Support Column (Solid)			
					Edition	Sheet 1 / 1

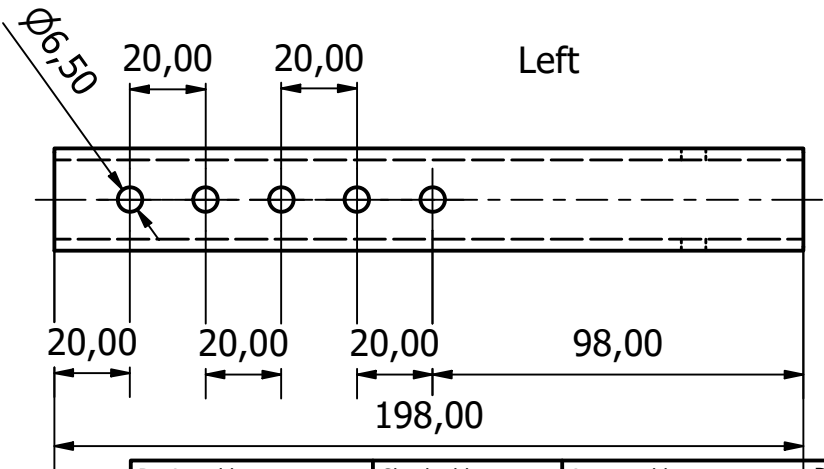
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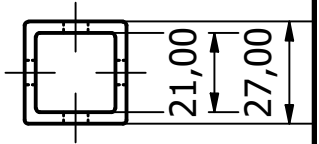
Isometric



Top



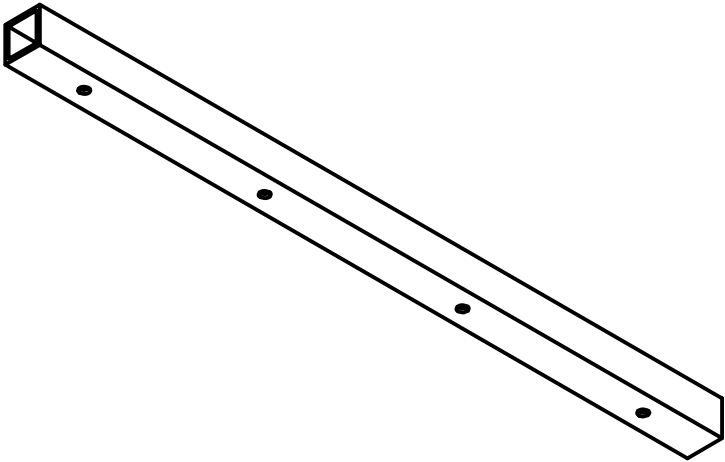
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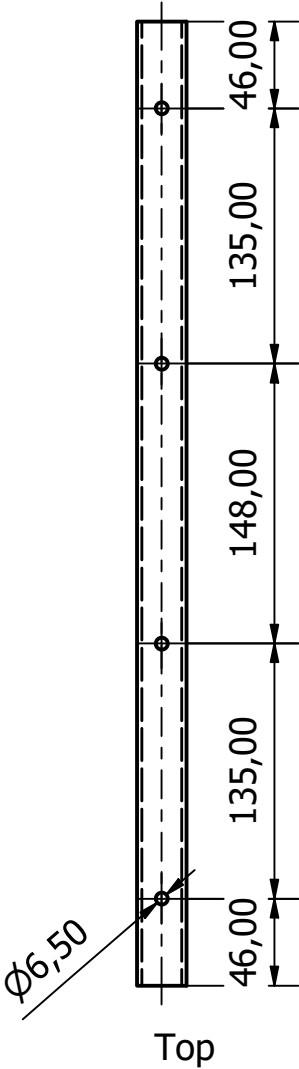
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Designed by loveraj	Checked by	Approved by	Date		Date	
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					Edition	Sheet 1 / 1

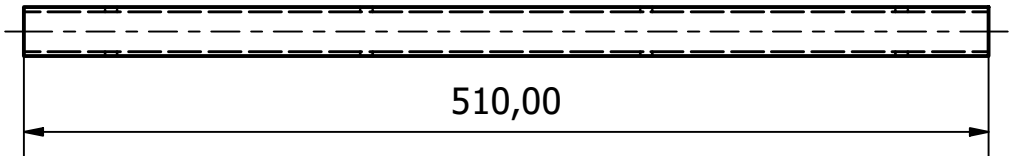
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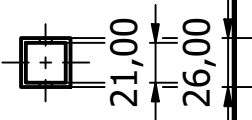
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Top



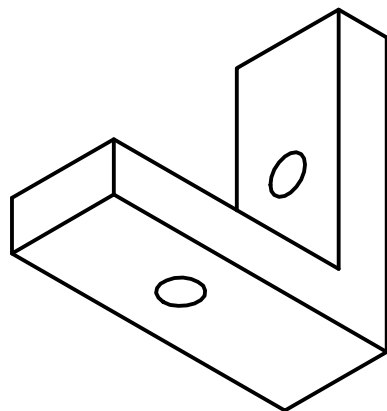
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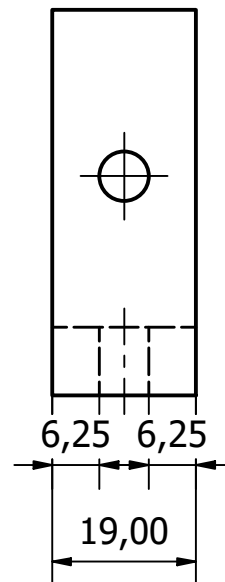
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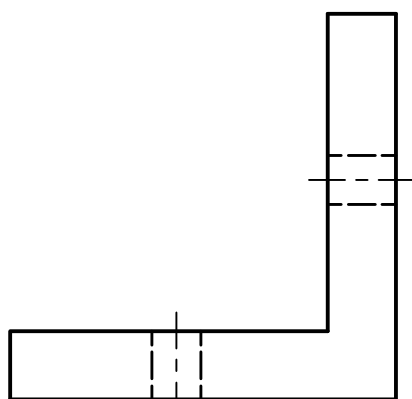
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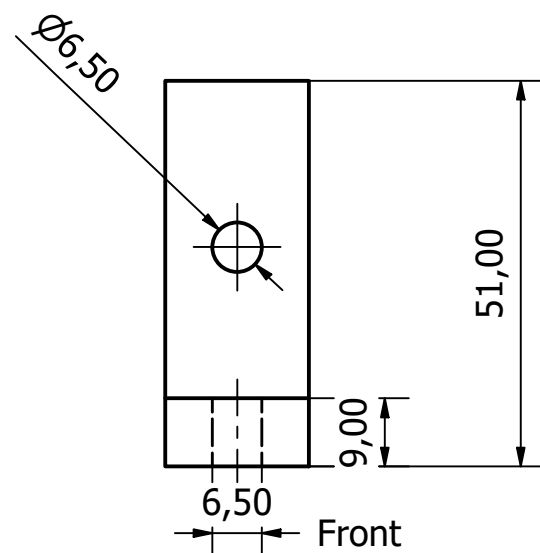
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Top

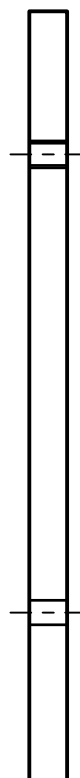
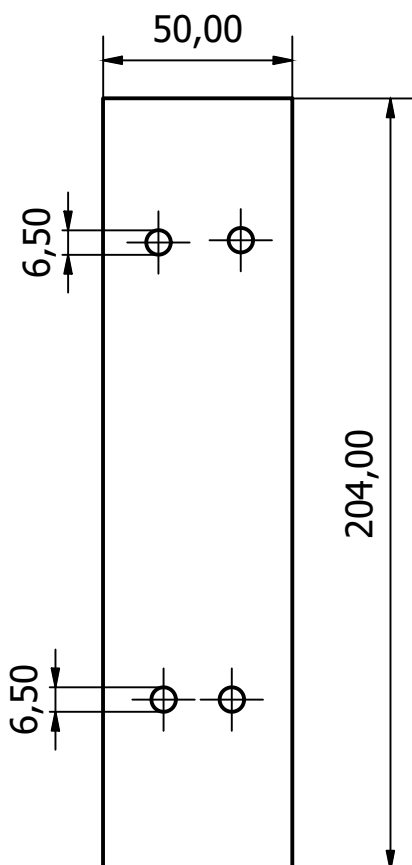
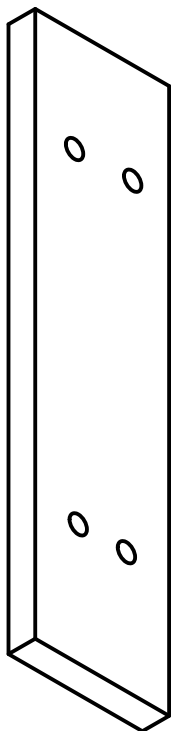


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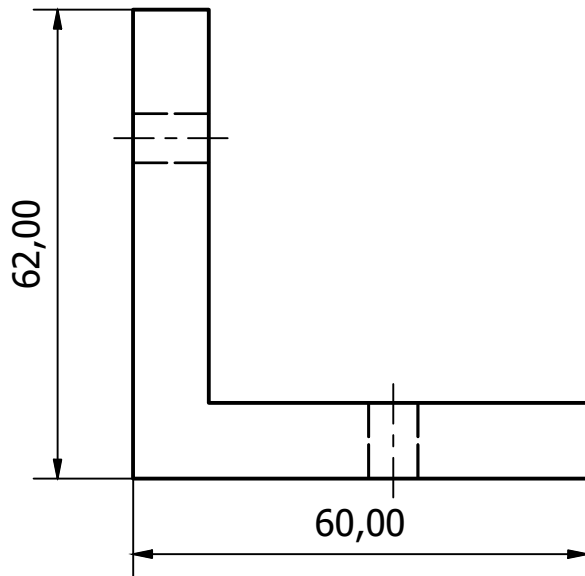
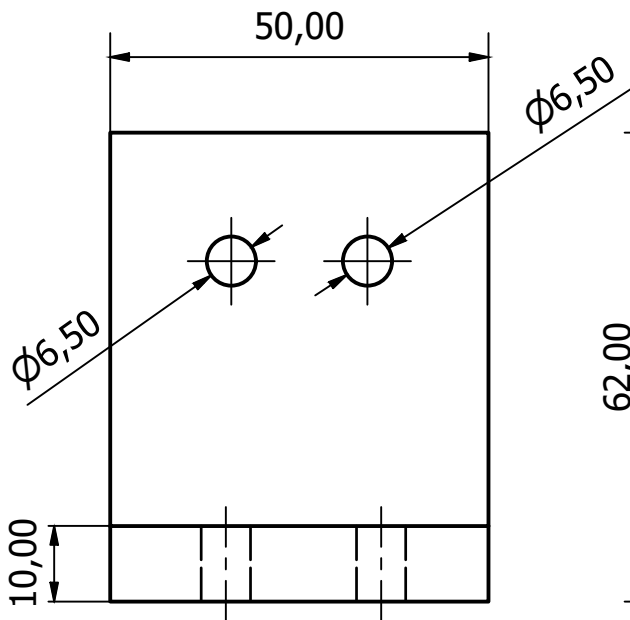
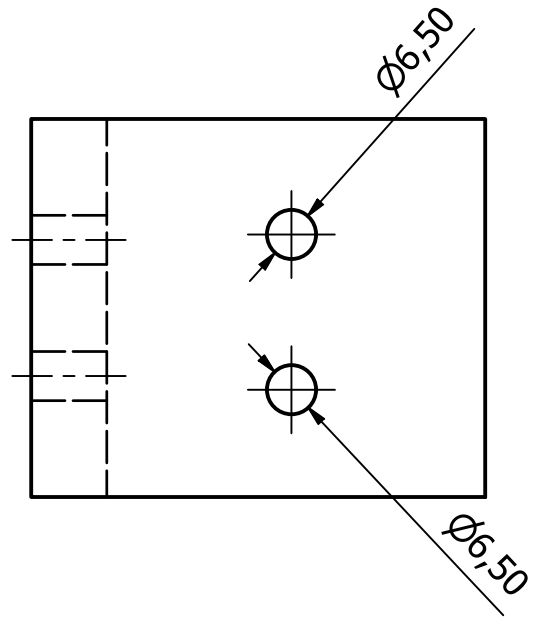
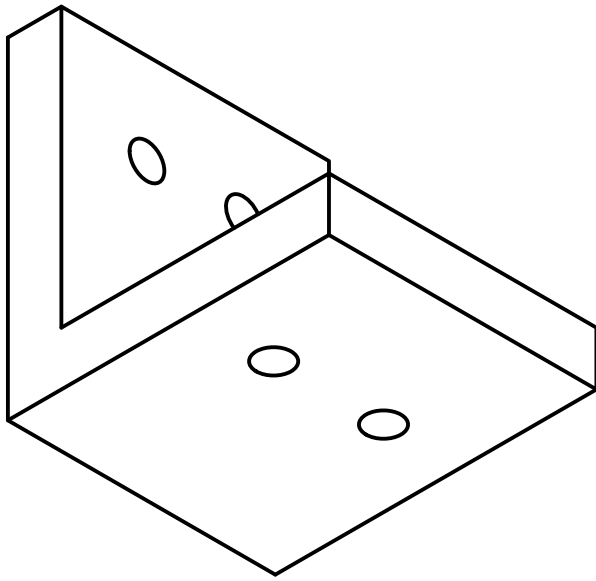


Front

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			Bottom Clamp Plate		
			Edition		Sheet 1 / 1

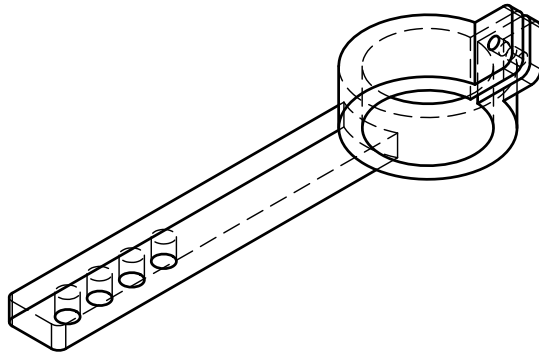


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Main Plate of Page Lifter arm Materials used: Mild Steel					
				Edition	Sheet 1 / 1

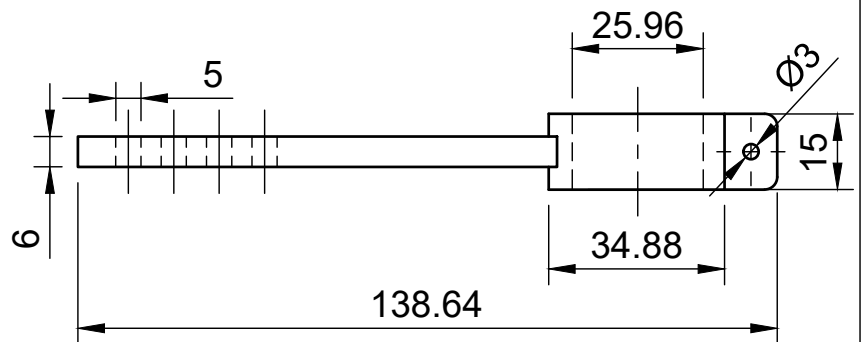
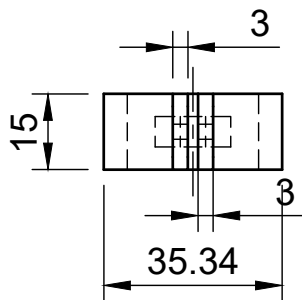
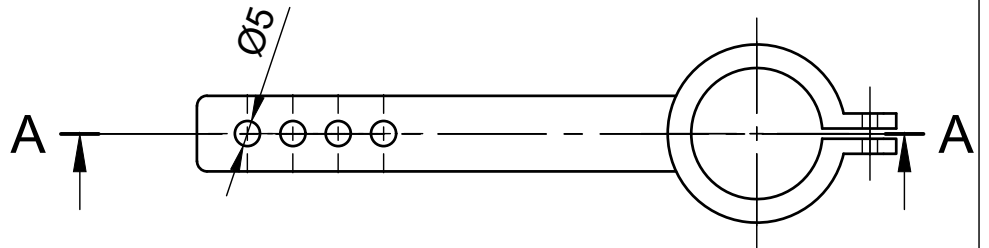
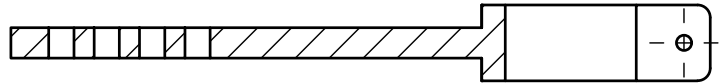


All measurements are in mm

Designed by Shivanshu_Jaiswara	Checked by	Approved by	Date	Date 30-10-2025	
Page Lifter Arm Support Materials used: Mild Steel					
				Edition	Sheet 1 / 1

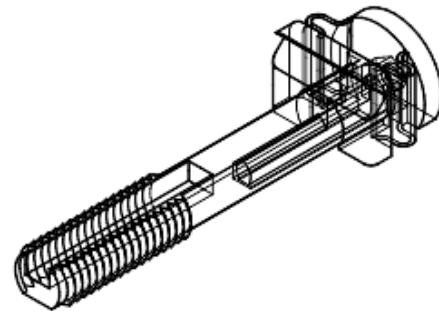
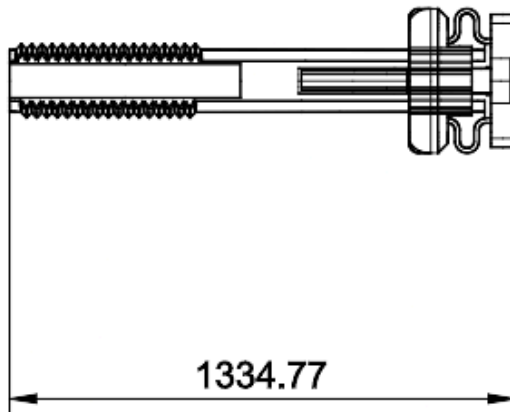
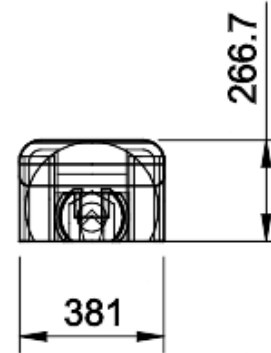
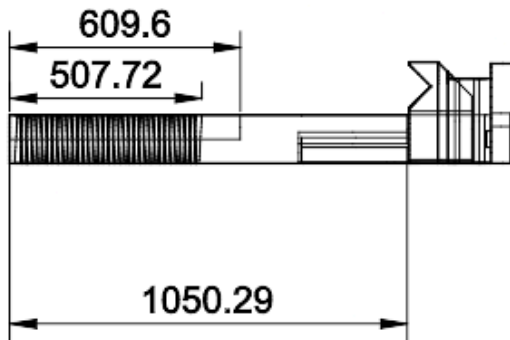


A-A (1:1.5)



All Measurements are in mm

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		Materials used: 3D Printed	Rev.	Date of issue Oct30, 2025	Sheet 1/1

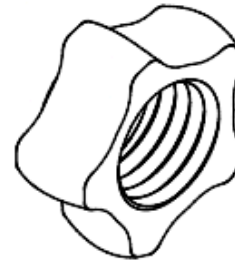
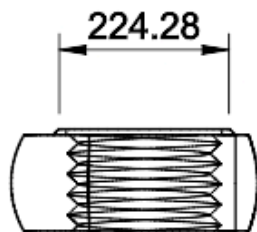
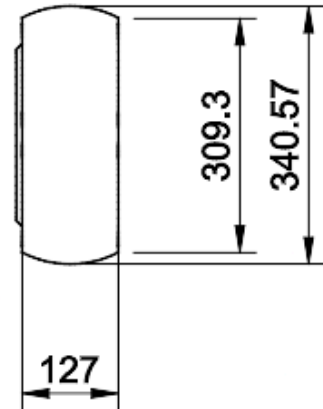
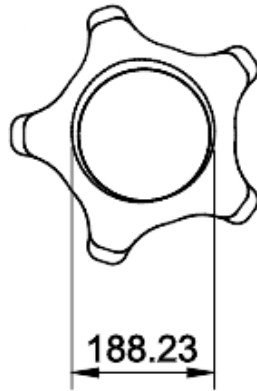


All dimensions are in mm

Material : PLA (poly lactic acid)
3D printed

Scale 1:20

Dept.	Technical reference	Created by Harshil Goel & Namya Sheth	Approved by		
		Document type	Document status		
		Title Mobile stand base	DWG No.		
			Rev.	Date of issue 30/10/25	Sheet 1/1

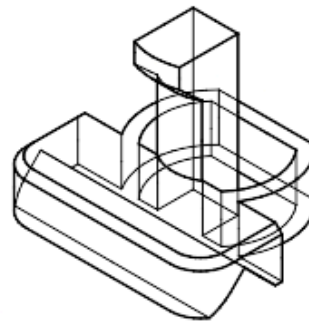
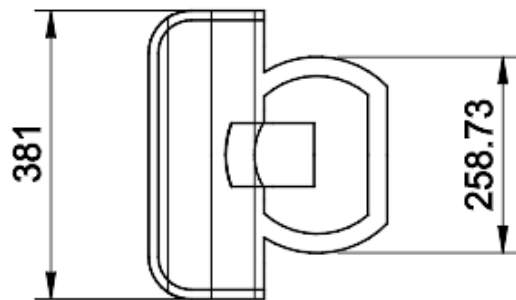
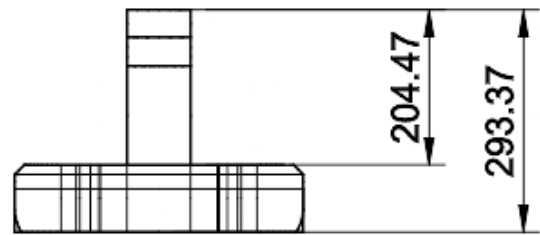
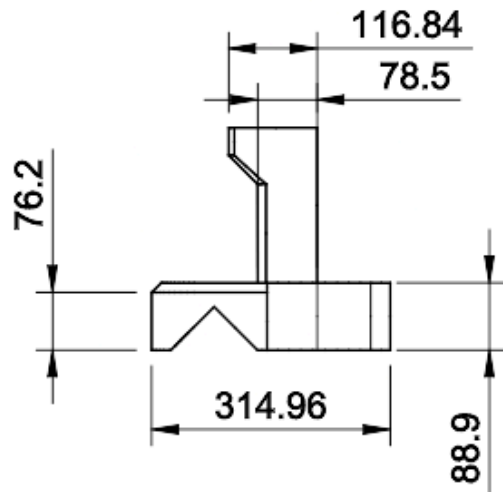


All dimensions are in mm

Material : PLA (polylactic acid)
3D printed

Scale 1:10

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		Document type	Document status		
		Title Mobile stand knob	DWG No.		
			Rev.	Date of issue 30/10/25	Sheet 1/1

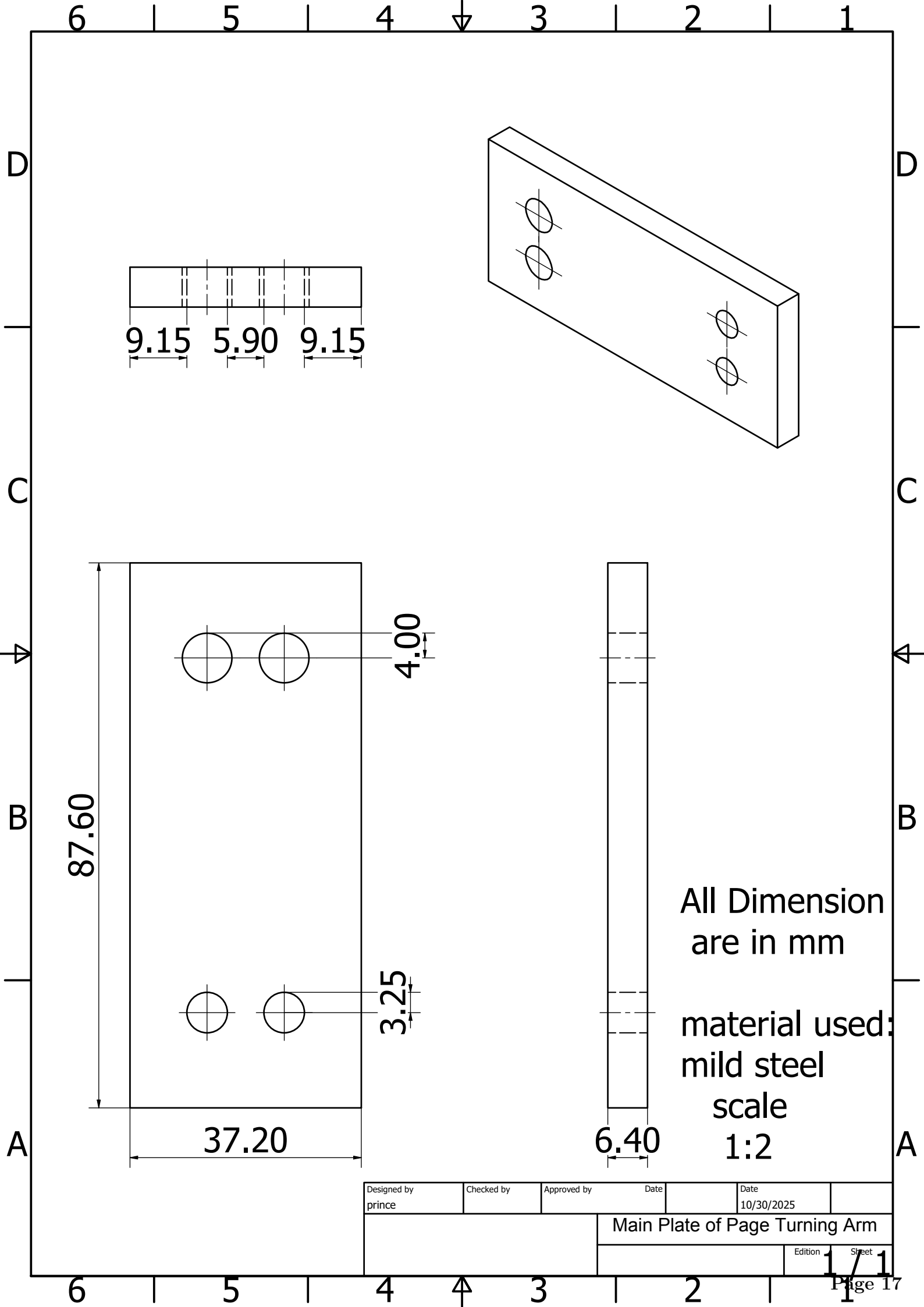


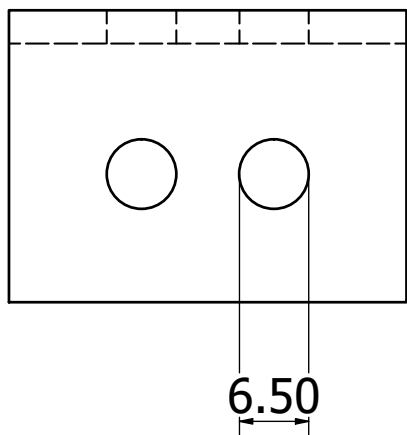
All dimensions are in mm

Material : PLA (polylactic acid)

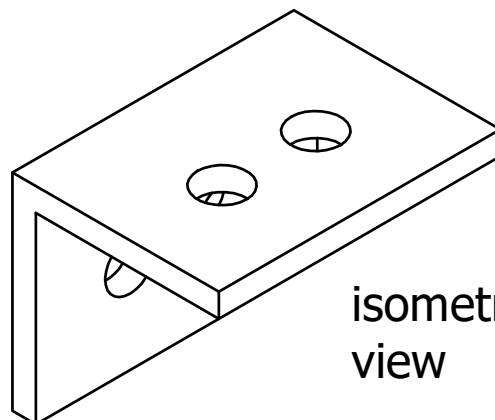
Scale 1:10

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			Rev.	Date of issue 30/10/25	Sheet 1/1





Top View

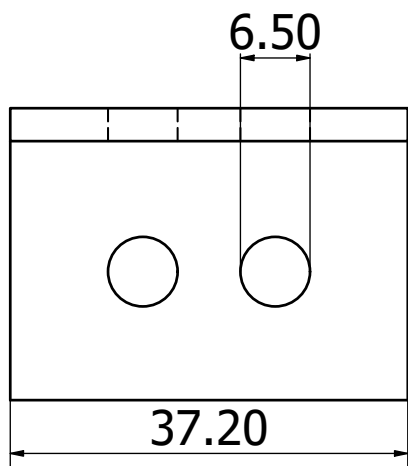


isometric
view

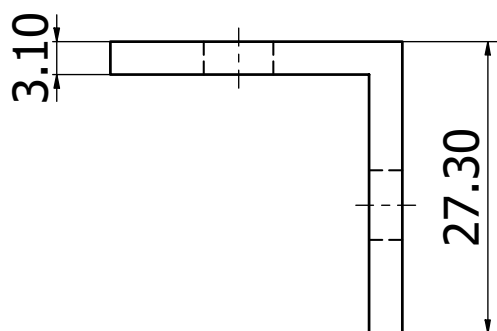
All dimensions are in mm

material used:mild steel

SCALE
1:1



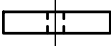
Front view



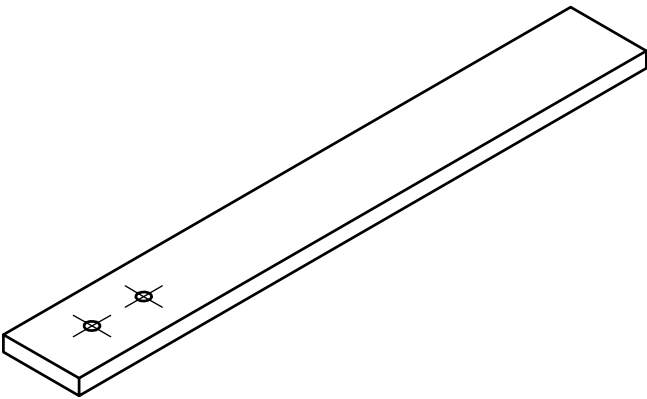
side view

Designed by prince	Checked by	Approved by	Date 10/30/2025	
			Servo motor's L-shaped clamp	
			Edition	Sheet 1 / 1

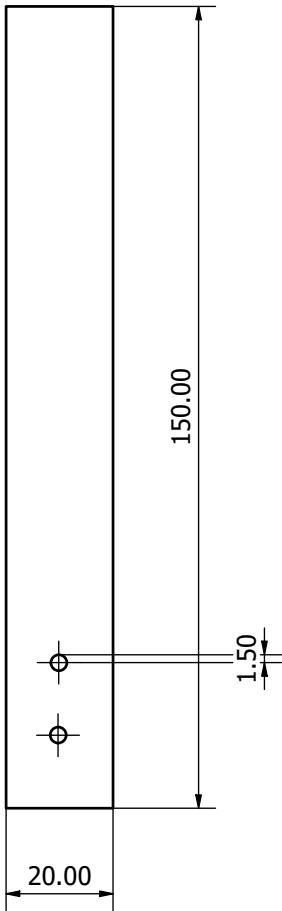
Page Turner



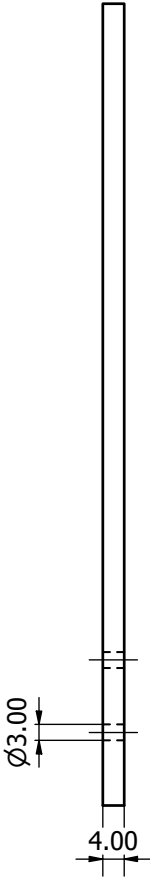
Top view



Isometric View



Front view



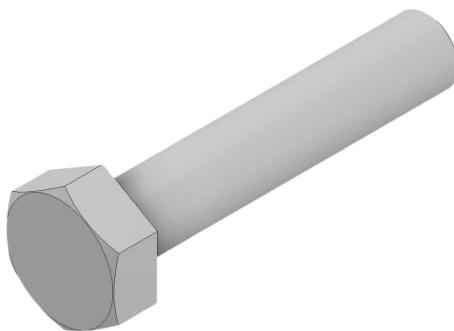
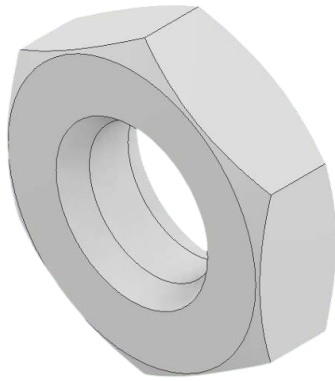
side view

Scale
1:1

Material used:
Polylactic acid
(All dimensions are in mm)

Designed by prince	Checked by	Approved by	Date	Date 10/31/2025	
Page Turning Arm					
				Edition	Sheet 1 / 1

Nut and Bolt



CODE IMPLEMENTATION

Github link of the page lifter and turning arm code of Automatic Book Digitaliser:

https://github.com/NamC-00/ta_proj

INTRODUCTION

Problem Statement

- Many **ancient scriptures and rare books** require preservation and wider accessibility.
- Manual **scanning of these fragile pages** can cause damage and is a time-consuming process.
- Traditional methods also demand **continuous human effort**, making the digitization of large volumes inefficient.

Proposed Solution

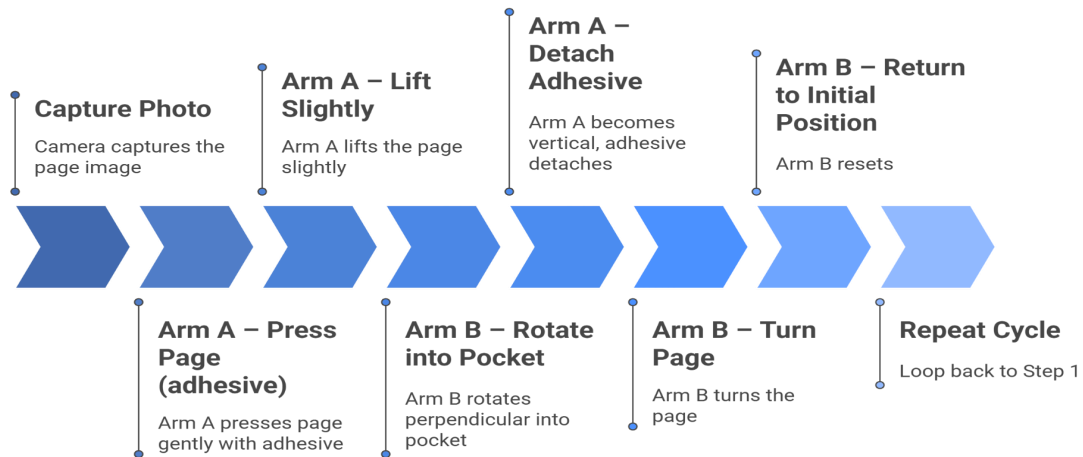
- We propose an **Automatic Book Digitaliser**, a system that automates page turning and scanning.
- The setup uses a mechanical arm with controlled motion to lift and turn pages, while a mounted camera captures each page.
- The scanned images are then **automatically converted into a PDF**, which can be uploaded for public access, digital libraries, and archival purposes.
- This automation **reduces human involvement, improves accuracy, and preserves** delicate texts safely.

Working of the Model

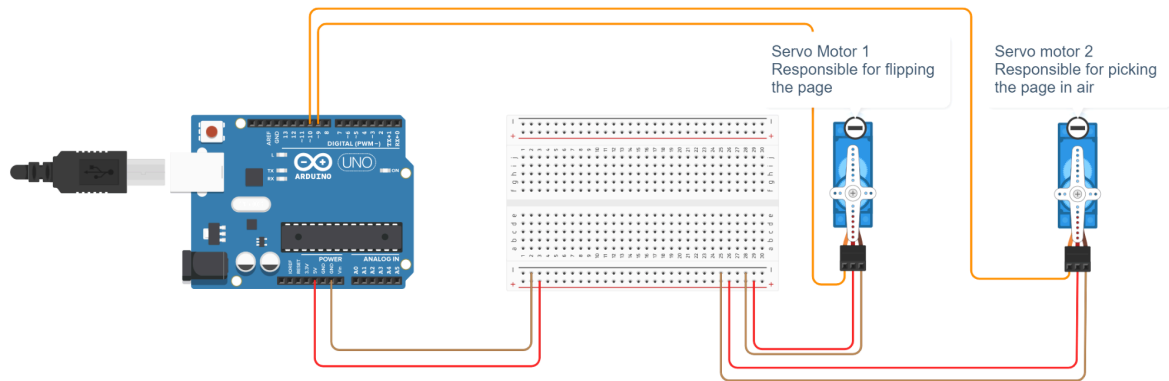
- An axle mounted on an L-plate moves vertically to align with each page.
- A small adhesive or suction-based tip grips and slightly lifts the page.
- A slider mechanism moves below the lifted page to flip it to the next side.
- A camera or smartphone mounted above captures the image after each page turn.
- Captured images are processed and compiled into a single PDF document.
- The entire process is automated, making book scanning faster, safer, and more efficient.

Working Flow of the model

Automatic Book Digitalizer – Page Turning Mechanism



ELECTRONICS



- *Power Supply and Common Ground*

The Arduino UNO is powered through the **USB connection** from the laptop, which also provides serial communication for data exchange. The breadboard power rails are supplied from the Arduino's 5V and GND pins. A common ground is maintained across all components — Arduino, servos to ensure stable and reliable signal referencing.

- *Servo Motor Connections*

Servo 1 (Page Turning Arm):

Signal pin → Digital Pin 9 (D9) on Arduino.

Vcc → +5V rail of the breadboard.

GND → Common Ground rail.

Servo 2 (Page Lifting Arm):

Signal pin → Digital Pin 10 (D10) on Arduino.

Vcc → +5V rail.

GND → Common Ground rail.

Each servo receives smooth PWM control signals from the Arduino to perform coordinated page-turning and lifting movements.

- *Communication and Control*

- The Arduino UNO communicates with the laptop via the USB serial port.
- When the Arduino completes one page-turning cycle, it sends a “**CAPTURE**” signal through the serial interface.
- A Python program running on the laptop listens for this serial message and automatically triggers a camera click event.
- The laptop sends this trigger command to a mobile phone via Bluetooth, which captures the image of the current page.

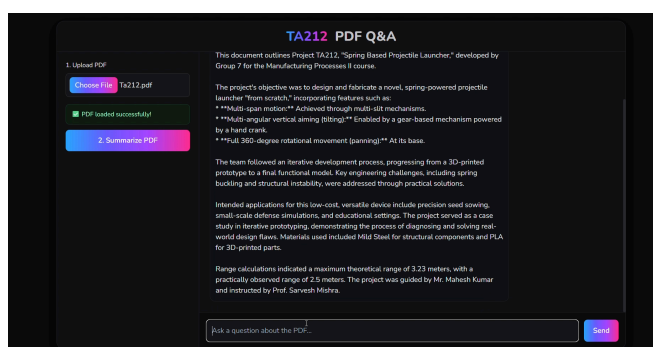
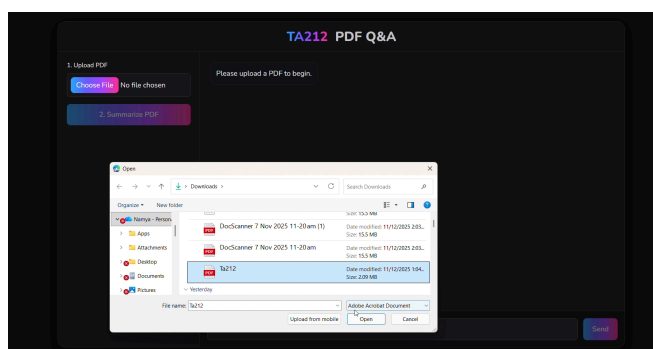
● *System Overview*

The entire system operates as an **integrated electromechanical setup** where:

1. Arduino controls the mechanical movement (page lift + turn).
2. The laptop handles communication and automation through Python.
3. The mobile phone captures images wirelessly via Bluetooth trigger.

This integration enables automatic, synchronized page turning and image capture for book digitization.

Website Demo



ACKNOWLEDGEMENT

We would like to convey our heartfelt gratitude to *Dr. Sarvesh Mishra*, our Course Instructor, for giving us the opportunity to undertake this project as part of the TA212: Manufacturing Processes II course. His valuable guidance, encouragement, and insightful feedback throughout the project were crucial in enhancing our understanding and approach.

We are deeply thankful to our Guide, *Mr. Abhishek Kumar*, for his continuous support, motivation, and technical guidance. His suggestions and supervision played a vital role in the smooth execution and successful completion of this project.

We would also like to acknowledge our Lab In-Charge, *Aman Singh*, for his assistance and cooperation. Finally, we extend our sincere thanks to our classmates, laboratory staff, and everyone who directly or indirectly contributed to the success of this project.

ABSTRACT

This project presents a **cost-effective, automated system** developed to rapidly digitize bound physical books and documents with minimal human intervention.

The system employs a two-arm **mechanical page-turning mechanism** driven by servo motors. The mechanism utilizes a Page Lifter arm with reusable adhesive to separate pages, and a Page Turning arm to flip the page over the book spine's center of gravity, ensuring gentle and precise page handling. The operation is controlled by an **Arduino**

Microcontroller, which executes a timed, closed-loop cycle of page turning, settling, and image capture.

Image capture is performed by a fixed-position camera (e.g., a mirrored mobile phone) remotely triggered by a Python script on a connected PC. This integration of mechanics and electronics creates a **repeatable and consistent apparatus** for unattended batch scanning, addressing critical needs such as the digitalization of ancient scriptures and mandatory digital health records.

MOTIVATION

The primary motivation for developing the Automatic Book Digitalizer is to provide an efficient and easy solution for the increasing demand for digital conversion of physical records. The project specifically aims to solve several real-world digitization problems:

- *Digitalization of Ancient Scriptures:*

The system enables the digitalization of very old, sacred, or culturally important books and documents with minimal human effort. This preserves cultural heritage and allows these works to be shared globally.

- *Digital Health Records Mandate:*

The project addresses the increased demand for digital conversion driven by the "Digital Health Records (Mandatory Use)" bill passed on 7th February, 2025, which mandates the use of digital health records in government healthcare institutions.

- *School and College Records:*

The digitalizer is positioned as a solution for converting extensive school and college records into digital formats.

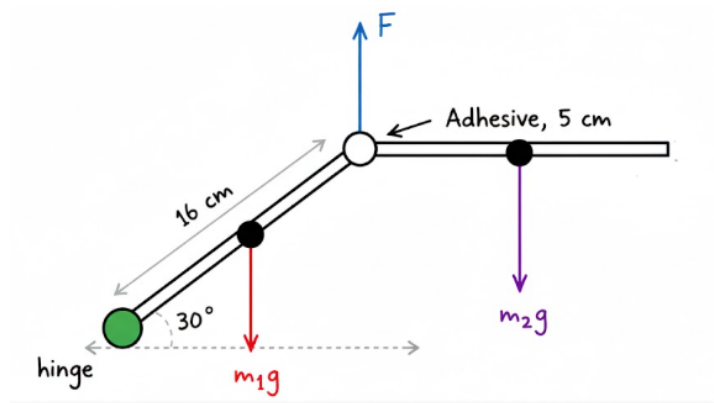
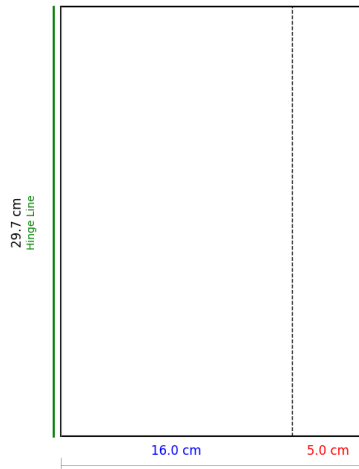
- *Minor Inconveniences:*

It is intended to simplify and automate general book scanning, which typically involves manual page turning.

TORQUE CALCULATION

Sticking strength of adhesive: We assume with dimensions 29.7cm x 21 cm and GSM(grams per square meter) to be 90 g/m^2

Top View of the Page without Adhesive
Dotted line is where the adhesive tape will be placed



Here,

M_1 is the mass of the part of the page that is 16 cm wide

M_2 is the mass of the part of the page that is 5 cm wide

We calculate the force exerted by the adhesive by performing torque analysis is equilibrium about the hinge of the book.

First let's calculate the mass of the page.

$$M = (\text{GSM}) \times (\text{Area of Sheet}) = 90 \times 10^{-3} \times 0.21 \times 0.297 = 5.6 \times 10^{-3} \text{ kg}$$

$$M_1 = M \times 16 \text{ cm} / 21 \text{ cm} = 4.27 \times 10^{-3} \text{ kg}$$

$$M_2 = M \times 5 \text{ cm} / 21 \text{ cm} = 1.33 \times 10^{-3} \text{ kg}$$

Using Newton's Second Law of Motion for moments of forces :

$$\sum \tau_i = 0 ; \text{ when } \tau_i \text{ represents the torque about the hinge.}$$

So we get:

$$F \times 16 \cos 30^\circ = M_1 g [(16 / 2 \times \cos 30^\circ) + M_2 g [(5 / 2 + 16 \cos 30^\circ)]$$

Assuming $g = 9.8 \text{ m/s}^2$ and by putting all the values,

$$F = 0.0363 \text{ N}$$

This is equivalent to holding a mass equal to F / g .

$$F / g = 0.0037 \text{ kg} = 3.7 \text{ g}$$

In testing we found that our adhesive can easily hold at least 5 grams, so there is no possibility of the page detaching from the adhesive.

Torque required by motor :

Torque required by the motor is $\tau = F \times r$, where r is the hinge length, equal to 15 cm.

We need to find the maximum value of F needed to lift the page during any point of its motion.

So,

$$F \times 2 \sin \theta \geq m \times g \times 10.5 \times \cos \theta$$

$$F \geq (m \times g \times \cot \theta) / 2$$

θ varies from 30° to 90° , since after 90° , the page falls down on its own.

Maximum value of $\cot \theta$ in the range is $\cot 30^\circ = \sqrt{3}$

$$\tau \geq 0.5007 \text{ N}$$

$$\tau = F \times r \geq 0.5007 \times 15 = 7.51 \text{ N.cm}$$

The units of kg.cm , this is equal to $7.51 / 9.81 \text{ kg.cm} = 0.766 \text{ kg.cm}$

We get 1.5 kg.cm at 4.8 V

The motor used in the project operates at 5V

Our motor can easily provide 1.5 kg.cm and thus calculation is as estimated above.

COST ESTIMATION

SR NO.	PART NAME	APPROXIMATE COST
1	Main base	Rs. 1,034.97
2	Book base	Rs.178.04
3	Book base support rods	Rs. 54.69
4	Vertical Support Column (Solid)	Rs. 75.35
5	Vertical Support Column (Hollow)	Rs. 55.95
6	Top Cross Beam	Rs. 243.57
7	Bottom Clamp Plate	Rs. 139.66
8	Main Plate of Page Lifter arm	Rs. 72.06
9	Page Lifter Arm Support	Rs. 131.41
10	Main Plate of Page Turning Arm	Rs. 14.73
11	Servo motor's L-shaped clamp	Rs. 2.22
12	Nut and Bolt	Rs. 48.62
13	Reusable Adhesive	Rs. 125

SR NO.	MACHINE	HOUR/S USED	APPROXIMATE COST
1	Lathe	2	Rs.700
2	Drilling	4	Rs.400
3	Cutting	2	Rs.120
4	3D printing	3	Rs.900

SR NO.	ELECTRONICS	COST
1	Arduino uno	Rs. 2000
2	Breadboard	Rs. 150
3	Jumper cables	Rs. 15
4	Servo Motors	Rs. 180

SR NO.	LABOUR	Hour/s Used	COST
1	Unskilled Labour	8	Rs. 650
2	Skilled Labour	8	Rs. 850

Note - For cost estimation purposes, the time utilized in machining processes has been taken as skilled labour hours, while the remaining activities such as assembly, cutting, and finishing are considered to fall under unskilled labour work.

Following cost approximations for the cost analysis of the mechanical prototype has been made :

Material	Cost / Kg.
Mild steel	Rs. 90/Kg
Nuts bolts	Rs 130/ Kg

Machine usage Cost:

3D Printing	Rs. 300/Hrs.
Lathe	Rs. 350/Hrs.
Drilling	Rs. 100/Hrs.
Cutting	Rs. 60/Hrs.

Labor Cost

Unskilled	Rs. 650/Day/8 Hrs.
Skilled	Rs. 850/Day/8 Hrs.

Final Estimated Cost

- Mechanical parts: Rs. 2,176.27
- Fabrication / machine work: Rs. 2,120.00
- Electronics: Rs. 2,345.00
- Labour: Rs. 1500

Total Estimated Cost = Rs. 2,176.27 + Rs. 2,120.00 + Rs. 2,345.00 + Rs. 1500 ≈ Rs. 8,141